

were computer-generated using individualized HRTFs measured from the entrance of the blocked ear canal (for reasons of repeatability and good signal-to-noise ratio⁹). A position-independent transformation was then applied to these blocked-canal HRTFs to make the pressure at the eardrum of the listener during playback consistent with the free-field pressure at that point. Natural-sound field stimuli were presented from a loudspeaker which was located 1.15 m from the listener's head. Both the real and the natural stimuli had a bandwidth of 350–15,000 Hz.

The presence of the 1.35-mm-diameter tubes in the ear canals had negligible effects on the natural sound field. This was observed subjectively, and confirmed by acoustical measurements and by a study comparing localization performance in the median-sagittal plane with and without the tube-phones present. We also measured the acoustical crosstalk between the tube-phones at the two ears to be negligible at the presentation levels used (50 dB sound pressure level (SPL) measured with steady-state excitation in the ear canal of the KEMAR acoustical mannequin).

Psychophysical discrimination paradigm. The experimental paradigm was a two-interval, two-alternative, forced-choice (2I, 2AFC) discrimination task. Each interval consisted of a pair of 80-ms noise bursts (having 10-ms raised-cosine rise/fall times) presented at 45 dB SPL and separated by 100 ms. The time between the two intervals of each trial was 200 ms. The stimuli in each interval were either synthesized virtually or delivered from a free-field loudspeaker in an anechoic chamber (from the corresponding location), in random order. In the experiments in which the HRTF spectrum was smoothed, the signal components of the noise stimuli were randomized in 1/3-octave bands by ± 5 dB in each trial to prevent listeners from using any non-spatial differences in sound quality in discriminating the virtual sounds from the real sounds. In a localization experiment conducted in the median-sagittal plane we determined that the localization performance of listeners was not significantly different with or without the spectral rove in the noise stimulus.

The location of a single source was tested during each run and feedback regarding the correct answer was provided in each trial. HRTFs were measured from listeners at the start of each experimental run with head position monitored by a Polhemus head-position sensor (which was worn on top of the subject's head, held by a plastic frame). A computer program instructed the subject to find 0-degree yaw, pitch and roll coordinates; the program read the Polhemus sensor output and delivered voiced instructions (consisting of digitized speech samples) over a loudspeaker. The instructions consisted of the words 'turn', 'roll', 'left', 'right', 'up', 'down', and 'hold it'. The measurement was only made when the reference position had been reached. The subject remained seated in the same location for the rest of the session and was prompted (if necessary) to achieve the same orientation for each presentation during playback. Head position was constantly monitored during the course of the stimulus playback and the trial was automatically terminated if any motion was detected. This ensured that discrimination between the stimuli did not result from any misalignment between the listener and the speaker and that no dynamical cues were used to determine the source of the stimulation.

HRTF smoothing. The discrete Fourier transform, $H(e^{j\omega})$, of an HRTF impulse response can be expressed as a discrete sequence, $H[k]$, where $H[k] = H(e^{j\omega})|_{\omega = (2\pi/N)k}$, where N is the number of sample values. The discrete realization of the log-magnitude spectrum may be expressed by the Fourier series:

$$\hat{H}[k] = \log|H[k]| = \sum_{n=0}^{N/2} C(n)\cos(2\pi nk/N)$$

where $C(n)$ is the n th coefficient in the Fourier series for the sequence $\log|H[k]|$.

By using a partial Fourier series of degree $M < N/2$, a smoothed fit to the HRTF spectrum can be obtained:

$$\hat{H}[k] = \sum_{n=0}^M C(n)\cos(2\pi nk/N)$$

The choice of $M = N/2$ corresponds to an exact reconstruction of the HRTF magnitude spectrum and the choice of $M = 1$ corresponds to a reconstruction having a flat spectrum at the average value of the HRTF.

For an empirical HRTF impulse response having N sample values, a Fourier series having $M = N/2$ terms provides an exact reconstruction of the empirical magnitude spectrum. The empirical HRTF in our experiments consisted of

1,024 time points (and thus correspond to a Fourier series with 512 terms). The smoothed magnitude spectra used in our experiments were constructed using 256, 128, 64, 32, 16 and 8 Fourier coefficients. The HRTFs were implemented as minimum-phase filters, augmented with a frequency-independent time delay, which was consistent with the overall ITD in the empirical HRTF measurements for the location being tested.

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Hedgehog stimulates maturation of *Cubitus interruptus* into a labile transcriptional activator

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In *Drosophila*, signalling by the protein Hedgehog (Hh) alters the activity of the transcription factor *Cubitus interruptus* (Ci) by inhibiting the proteolysis of full-length Ci (Ci-155) to its shortened Ci-75 form^{1,2}. Ci-75 is found largely in the nucleus and is thought to be a transcriptional repressor¹, whereas there is evidence^{3–5} to indicate that Ci-155 may be a transcriptional activator^{1,2,6}. However, Ci-155 is detected only in the cytoplasm, where it is associated with the protein kinase Fused (Fu), with Suppressor of Fused (Su(fu)), and with the microtubule-binding protein Costal-2 (refs 1,7–9). It is not clear how Ci-155 might become a nuclear activator. We show here that mutations in *Su(fu)* cause an increase in the expression of Hh-target genes in a dose-dependent manner while simultaneously reducing Ci-155 concentration by some mechanism other than proteolysis to Ci-75. Conversely, eliminating Fu kinase activity reduces Hh-target gene expression while increasing Ci-155 concentration. We propose that Fu kinase activity is required for Hh to stimulate the maturation of Ci-155 into a short-lived nuclear transcriptional activator and that *Su(fu)* opposes this maturation step through a stoichiometric interaction with Ci-155.

Hh secreted from posterior compartment cells of the wing disc in *Drosophila* induces expression of *decapentaplegic* (*dpp*) in anterior cells as far as the site of the future vein 3 (refs 10, 11) and anterior *engrailed* (*en*) expression in a narrower strip of cells close to the anterior–posterior (AP) compartment border^{11,12} (Fig. 1a). The amount of Dpp protein secreted from the AP border dictates the position at which vein 2 forms¹³. En and other factors induced by Hh contribute to the formation of vein 3 and associated campaniform sensillae at the anterior edge of the Hh-signalling territory¹⁴.

These effects of Hh signalling can be reproduced in more anterior regions by ectopic expression of Hh¹⁰, or by genetic inactivation of the Hh-receptor Patched (Ptc) in somatic clones² (Fig. 1a–c; Table 1). Anterior clones deficient in protein kinase A (PKA) activity also induce ectopic expression of Hh-target genes and reorganize wing pattern² (Fig. 1e; Table 1).

Elimination of *Su(fu)* activity moderately increased the effects of *ptc* and *PKA* mutant clones, enhancing the displacement of vein 2 from the anterior edge of the mutant clone and the number of ectopic sensillae induced (Table 1). A similar increase in sensillae and the distance between vein 2 and vein 3 was produced by *Su(fu)* mutations in otherwise wild-type wings, indicating a possible effect on normal Hh signalling (Table 1). Transcriptional responses were also affected. Ectopic *En* expression was induced less strongly in anterior *PKA* mutant clones than in *ptc* mutant clones (Fig. 2a, g), but was increased by loss of *Su(fu)* only in *PKA* mutant clones to reach the same level as in *ptc* clones (Fig. 2c, i).

A more marked enhancement of *PKA* mutant phenotypes by *Su(fu)* mutation was seen in embryos. In stage-9–12 embryos, *wingless* (*wg*) is normally expressed in one cell-wide stripes in each segment: ectopic Hh protein or *ptc* mutations induce an anterior expansion of the *wg* stripes^{5,15}, whereas inhibition of PKA causes

very little or no ectopic *wg* expression¹⁶. Loss of *Su(fu)* activity greatly enhanced induction of ectopic *wg* RNA by a dominant PKA inhibitor expressed in roughly alternating segments (Fig. 3a, c).

Hh signalling or loss of *ptc* or *PKA* activity are each accompanied by an increase in the amount of Ci-155 that can be detected *in situ* with an antibody that recognizes Ci-155 but not Ci-75 (refs 1, 2, 17) (Fig. 2b, h). Surprisingly, *Su(fu)* mutations decreased the amount of Ci-155 induced by Hh or by loss of *ptc* or *PKA* activity. Ci-155 staining at the AP border (Fig. 5b) was greatly reduced in *Su(fu)* wing discs (Fig. 5f). Ci-155 concentrations were only slightly increased in *PKA* mutant clones in *Su(fu)* discs (Fig. 2j) and no increase of Ci-155 was detected in *ptc* mutant clones in *Su(fu)* discs (Fig. 2d). Inhibition of PKA in embryos increased the amount of Ci-155 (ref. 16) (Fig. 3b), but not in the absence of *Su(fu)* activity (Fig. 3d).

Western blots of wild-type and *Su(fu)* mutant embryo and wing-disc extracts confirmed that *Su(fu)* mutations reduced the accumulation of Ci-155 (Fig. 4). Furthermore, PKA inhibition greatly increased the amount of Ci-155 at the expense of Ci-75 in embryos, whereas loss of both PKA and *Su(fu)* activity reduced Ci-155 to below the level in wild-type embryos. The reduction in Ci-155 caused by *Su(fu)* mutation was never accompanied by an increase in Ci-75, indicating that proteolysis of Ci-155 to Ci-75 was not responsible.

Conversion of Ci-155 to Ci-75 requires Slimb, which is thought to direct ubiquitin-mediated proteolysis of specifically phosphorylated forms of proteins, including Ci (ref. 18). There was much less Ci-155 in *slimb*² *Su(fu)* double-mutant clones (Fig. 5e) than in *slimb*² clones (Fig. 5a). Also, loss of *Su(fu)* reduced the accumulation of Ci-155 in *slimb*¹/*slimb*² wing discs, once again without increasing Ci-75 (Fig. 4). Thus, *Su(fu)* must affect Ci-155 levels by a mechanism other than the established Slimb-dependent conversion to Ci-75. The amount of *ci* RNA was not affected by *Su(fu)* mutation in whole discs (Fig. 5c, g). To account for the increased expression of Hh-target genes and the decrease in Ci-155 in *Su(fu)* mutants, we propose that the function of *Su(fu)* is to antagonize the conversion of Ci-155 to a more labile but transcriptionally active form in cells that are undergoing Hh signal transduction.

If *Su(fu)* blocks maturation of Ci-155 into an activator by binding to it⁷, then the concentration of Ci-155 relative to *Su(fu)* should affect the specific activity of Ci-155. Inactivation of one copy of the *ci* gene roughly halved the amount of Ci (Fig. 2n), but almost eliminated the effects of *PKA*-mutant clones on growth and patterning of wings. Normally, *PKA*-mutant clones are bordered on each side by an ectopic vein 3 and induce pattern duplications if located anywhere anterior to vein 3 (Fig. 1e). In *ci*/+ animals, *PKA*-mutant clones only induced additional anterior tissue if the clone formed anterior to vein 2 and in these cases induced an ectopic vein 3 in the middle of the clone (Fig. 1f). The normal organizing properties of *PKA*-mutant clones were restored in *ci* heterozygotes

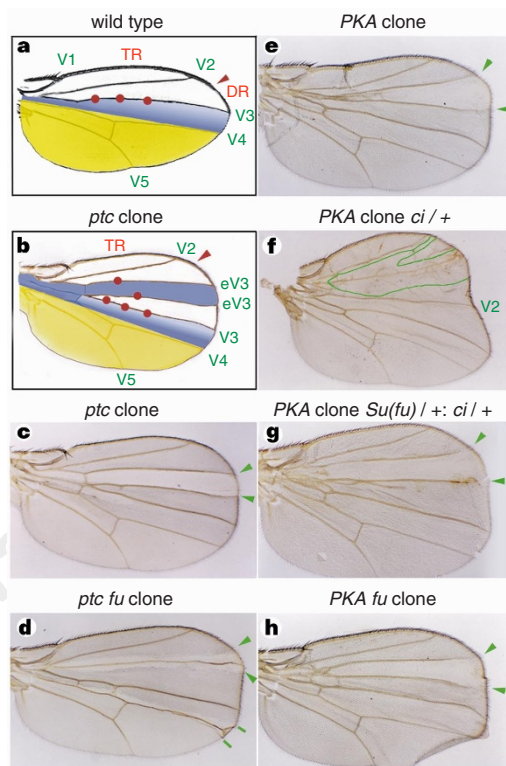


Figure 1 Effects of Ci/*Su(fu)* ratio and *Fu* kinase activity on wing patterning. **a**, Posterior compartment expression of Hh (yellow) and Hh-target genes, *dpp* (light blue) and *en* (dark blue), from third-instar wing discs projected on to a wild-type adult wing. At the edge of Hh signalling territory, vein 3 (V3), sensillae (red circles), and transition of marginal bristles to double row (DR) can be seen. **b**, Anterior *ptc* or *PKA* mutant clones between V3 and V2 induce ectopic Hh-target gene expression cell-autonomously (blue), ectopic vein 3 (eV3) and sensillae (red circles) either side of the clone, as well as anterior displacement of both V2 and the transition (arrowhead) to triple-row (TR) bristles. **c–i**, Mutant clones are seen between arrowheads and appear pale, except in **e**, where the clone is outlined in green. **c**, *ptc* clone; **d**, *ptc fu* double-mutant clone; **e**, *PKA* clone; **f**, *PKA* clone in *ci*/+ fly; **g**, *PKA* clone in *Su(fu)*/+; *ci*/+ fly; **h**, *PKA fu* double-mutant clone. **d**, A second *ptc fu* clone at the AP border (between the green lines) invaded the posterior, forming an ectopic vein 3 plus sensillae in place of vein 4, as did anterior *fu* clones at the AP border (not shown).

Table 1 *Su(fu)* and *Fu* modify wing phenotypes due to Hh signalling

		Distance to vein 2	Sensillae			Distance to vein 2	Sensillae
WT	F	41 ± 2	3.0	<i>Su(fu)</i>	F	46 ± 2	3.4
WT	M	40 ± 1	3.0	<i>Su(fu)</i>	M	45 ± 2	3.6
				<i>fu Su(fu)</i>	M	45 ± 2	3.2
<i>ptc</i> *	F	38 ± 1	0.5	<i>ptc Su(fu)</i> *	F	40 ± 2	2.1
<i>ptc</i>	M	38 ± 1	0.7				
<i>ptc fu</i> †	M	28 ± 3	0	<i>ptc fu Su(fu)</i> †	M	40 ± 2	2.6
<i>PKA</i> ‡	F	36 ± 2	0.1	<i>PKA Su(fu)</i> ‡	F	40 ± 2	2.0
<i>PKA</i>	M	33 ± 2	0				
<i>PKA fu</i>	M	32 ± 1	0.1	<i>PKA fu Su(fu)</i>	M	38 ± 3	2.3

Distances to vein2 from vein3 (WT) or anterior edge of mutant clone (*ptc*, *PKA*, *ptc fu*, *PKA fu*) are expressed as the sum of medial triple row and dorsal double-row bristles ± s.d. for females (F) and males (M). The number of sensillae distal to the vein 3–vein 4 crossvein that were on vein 3 (WT) or immediately anterior to mutant clones were counted. All values are means from at least 13 samples except for *ptc fu Su(fu)*.

* Also heterozygous for *fu* and *PKA*.

† Also heterozygous for *PKA*.

‡ Also heterozygous for *fu*.

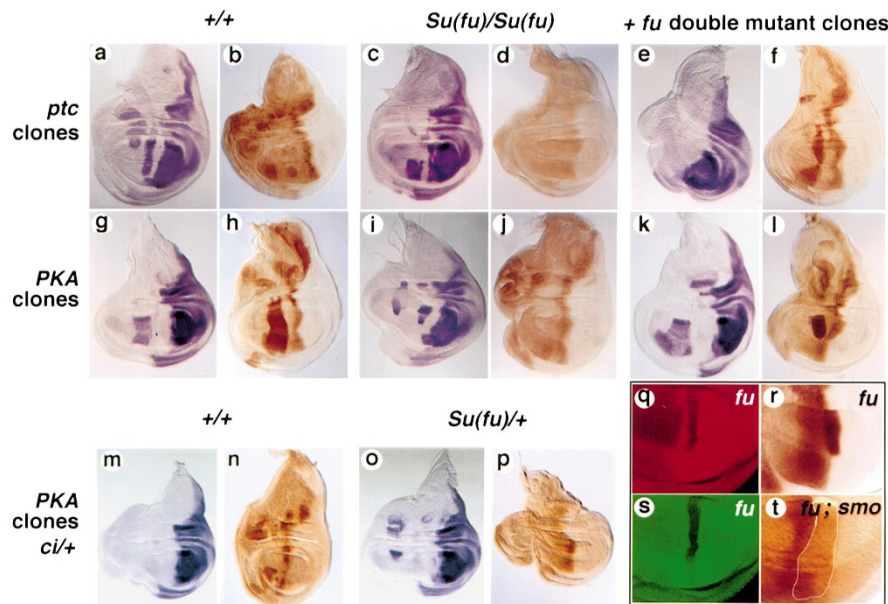


Figure 2 Effects of Ci/Su(fu) ratio and Fu kinase activity on anterior expression of En and Ci-155. **a–d**, *ptc* clones in discs with (+/+) or without *Su(fu)* activity; **g–j**, *PKA* clones in discs with (+/+) or without *Su(fu)* activity; **m–p**, *PKA* clones in *ci* heterozygotes that are wild-type (+/+) or heterozygous for *Su(fu)*; **e, f**, *ptc fu* double-mutant clones; **k, l**, *PKA fu* double-mutant clones; **q–s**, *fu* clones; **t**, *fu smo* double-mutant clone. En antibody staining (blue; red in **q**) was seen in the posterior (right) of all discs and at similar intensity in: **a**, *ptc*-mutant clones; and in **c**, *ptc*; or **i**, *PKA*-mutant clones in *Su(fu)/Su(fu)* animals. Less ectopic En was induced in: **g**, *PKA*-mutant clones; **k**, *PKA fu* double-mutant clones; and **o**, *PKA*-mutant clones in *Su(fu)/+*; *ci/+* animals. No ectopic En was induced in: **e**, *ptc fu* double-mutant clones; **m**, *PKA*-mutant clones in *ci/+* animals; or **q, fu** clones at the AP border. The presence of anterior clones in **e** and **m** is evident from disc distortions and ectopic Ci in sister discs (**f, n**). The *fu* clone in **q** is marked by the

absence of *arm-lacZ* staining (green) in **s**. **b**, Ci-155 stained with antibody 2A1 (brown) was exclusively anterior and was similarly increased at the AP border and in *ptc*-mutant clones. The highest ectopic expression of Ci was in: **h**, *PKA*- and **l**, *PKA fu*-mutant clones. There was more Ci than at the AP border in: **f**, *ptc fu* clones; and **r**, *fu* clones at the AP border, which projected into the posterior of the disc. **t**, *fu* clones also mutant for *smo* (outlined in white) projected into the posterior but stained only weakly for Ci-155 in the anterior, as seen for *smo* clones. In *Su(fu)/Su(fu)* animals, in **d** there was no increase in Ci-155 in *ptc* clones (disc distortion and ectopic En in sister discs (**c**) show the presence of clones); and in **j**, the amount of Ci-155 was only slightly raised in *PKA*-mutant clones. Ci staining in *PKA*-mutant clones was roughly halved in **n**, *ci/+* discs, and was even less in **p**, *Su(fu)/+*; *ci/+* discs.

by additional inactivation of one copy of the *Su(fu)* gene (Fig. 1g). Similarly, induction of En expression in *PKA*-mutant wing disc clones was lost in *ci* heterozygotes but was restored by loss of one copy of *Su(fu)* (Fig. 2g, m, o). Hence, the repressive effects of *Su(fu)* on Hh signal transduction depend on the relative concentrations of Ci and *Su(fu)*, consistent with a stoichiometric functional interaction.

The Fused protein appears to play a structural role in the cytoplasmic Ci-155 complex that helps to regulate proteolysis of Ci-155 to Ci-75 (ref. 8, J.T.O., unpublished results), but the function of its kinase activity has not been explained. A strong *fu* mutation, *fu*^{mH63}, which affects only the kinase domain, completely blocks transcriptional responses to Hh in the embryo but only partially in wing discs^{19,20}. Anterior wing-disc clones lacking both *ptc* and Fu kinase activity did not express ectopic En (Fig. 2e). The *ptc fu*^{mH63} clones induced ectopic sensillae and an ectopic vein 3 in the middle of the clone, and a much smaller displacement of vein 2 than *ptc* mutant clones (Fig. 1c, d; Table 1). Likewise, anterior *fu*^{mH63} clones at the AP border did not express En, invaded posterior compartment territory^{21,22} (Fig. 2q, s), and induced a central vein 3 decorated with sensillae at the position normally occupied by vein 4 (Fig. 1d; see legend).

Although it reduced the response to Hh and to *ptc* mutations in wing discs, the *fu*^{mH63} mutation increased the amount of Ci-155 at the AP border (Figs 2r; 5b, d) and in *ptc* mutant clones (Fig. 2b, f). Increased accumulation of Ci-155 was also evident in western blots of *fu*^{mH63} wing discs and was not accompanied by a decrease in Ci-75 (Fig. 4). Neither the level of Ci nor the wing morphology was affected by *fu*^{mH63} in mutant clones outside the Hh signalling region (not shown) or at the AP border when Hh signalling was blocked by

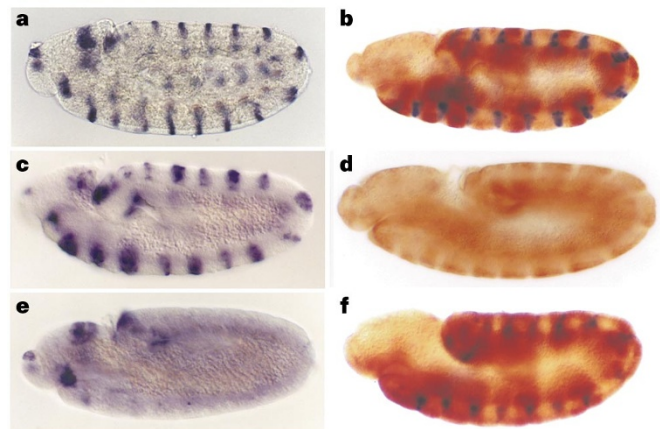


Figure 3 Loss of *Su(fu)* in embryos enhances ectopic *wg* expression and decreases Ci-155 levels resulting from PKA inhibition. **a, c, e**, *wg* RNA (blue); **b, d, f**, Ci-155 stained with antibody 2A1 (brown); in **b, f**, antibody staining for En (blue) is shown in stage-10–11 embryos. All embryos expressed the PKA-inhibitor transgene *UAS-R** under *prd-GAL4* control¹⁶. This led to: **a**, normal *wg* expression; and **b**, greatly increased Ci staining anterior to En stripes 1, 2, 3, 5, 7, 9, 11 and 13, where PKA is inhibited by *R** expression¹⁶. **c, d**, Embryos additionally lacking maternal and zygotic *Su(fu)* activity showed: **c**, expanded *wg* stripes 1, 2, 3, 5, 7, 9, 11 and 13, but **d**, no increase in Ci staining in segments where PKA was inhibited. **e, f**, Embryos additionally lacking maternal and zygotic Fu kinase activity: **e**, lost *wg* expression in trunk segments; and **f**, displayed increased Ci staining in regions of PKA inhibition. *fu* male embryos from germline clones were identified in **e**, by the absence of Sex-lethal antibody staining, and in **f**, according to decay of En expression.

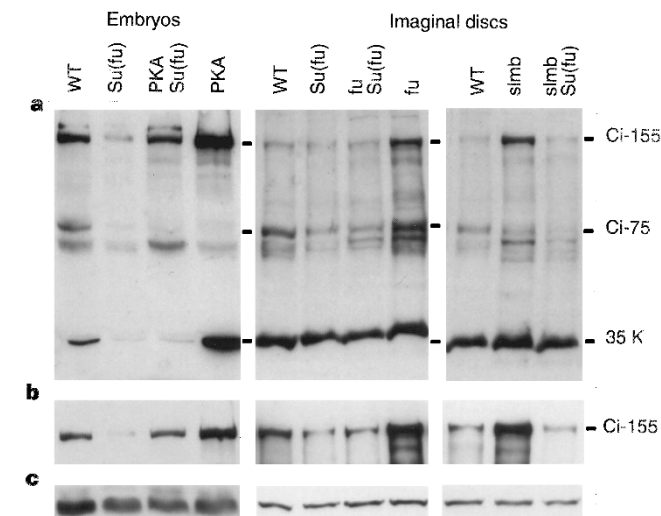


Figure 4 Western blots of embryo and wing disc extracts. **a**, Antibody raised against the zinc-finger domain of Ci recognizes Ci-155, Ci-75, a 35K band, and background bands smaller than Ci-75. **b**, Antibody 2A1 recognizes only Ci-155. **c**, Equal loading was determined by protein concentration ($\sim 20 \mu\text{g}$ protein per lane) and re-probing with antibody to the PKA type-II regulatory subunit (RII) for embryos, and according to disc number (25 discs per lane; 20 for *slimb* mutants) and re-probing with antibody to α -tubulin for wing discs. WT, wild type. In embryos, *PKA* denotes ubiquitous expression of *UAS-R** using *E22C-GAL4* (ref. 16) to inhibit PKA, and *Su(fu)* denotes loss of maternal and zygotic activity. In discs, *Su(fu)* and *fu* denote loss of zygotic *Su(fu)* and Fu kinase activity (*fu*^{*mH63*}). *slimb* denotes *slimb*¹/*slimb*² discs.

a null *smoothened* (*sno*) mutation^{23,24} (Fig. 2t). Hence, *fu*^{*mH63*} and *Su(fu)* mutations induced opposite effects on Hh-target gene expression and Ci levels that were in both cases limited to cells undergoing Hh signal transduction. In the absence of both *Su(fu)* and Fu kinase activity, the amount of Ci and the wing phenotype were typical of *Su(fu)* mutations²⁰ (Figs 4; 5h; Table 1). We suggest that Fu kinase normally opposes the action of *Su(fu)*, perhaps by promoting its dissociation from the Ci-155 complex.

This model predicts that Fu kinase activity is required only to activate Ci-155 that is associated with *Su(fu)*. The effects of Ci and *Su(fu)* dosage on *PKA*-mutant phenotypes in wing discs (Figs 1e–g; 2g, m, o) suggest that the amount of Ci-155 that is induced by *PKA* mutation overwhelms the inhibitory capacity of *Su(fu)*. If this excess Ci-155 is free of *Su(fu)*, it should be insensitive to Fu kinase. In agreement, we found that ectopic En and the wing phenotype induced by *PKA*-mutant clones was virtually unaffected by the *fu*^{*mH63*} mutation in *PKA fu*^{*mH63*} double-mutant clones (Figs 1e, h; 2g, k; Table 1). *ptc fu*^{*mH63*} double-mutant clones induced less Ci-155 than did *PKA fu*^{*mH63*} clones (Fig. 2f, l) but still retained some pattern-organizing activity in wings (Fig. 1d; Table 1), indicating that Ci-155 may be in excess relative to *Su(fu)*. By contrast, Fu kinase activity is essential for the induction of Hh-target genes in embryos, whether this is in response to Hh protein, *ptc* mutation⁵, or inhibition of PKA (Fig. 3e), even though PKA inhibition induced very high amounts of Ci-155 (Fig. 3f). We suggest that *Su(fu)* levels in embryos are comparatively high and are not exceeded by those of Ci-155 during Hh signalling or when PKA is inhibited.

Hh protein and *ptc* mutations elicit stronger phenotypic responses in embryos and wing discs than *PKA* mutations, despite the lower induction of Ci-155, indicating that Hh does more than simply inhibit the proteolysis of Ci-155 to Ci-75 (ref. 16). We have shown that the greater potency of Hh and *ptc* mutations relative to *PKA* mutations was suppressed or even reversed by eliminating *Su(fu)* or Fu kinase activity. We conclude that Hh promotes the Fu-kinase-dependent maturation of Ci into a transcriptional activator

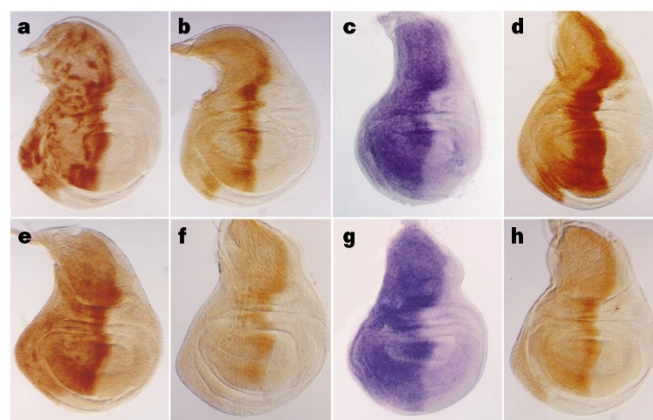


Figure 5 Effects of loss of *Su(fu)* on the level of Ci-155 are epistatic to *slimb* and *fu* mutations. **a, b, d, e, f, h**, Ci-155 staining with antibody 2A1 (brown): **a**, small *slimb*² anterior (left) clones induced high levels of Ci-155, whereas in **e**, Ci-155 was barely detectable above anterior levels in *slimb*² *Su(fu)* double-mutant clones; **b**, wild-type wing disc shows increased Ci staining at the AP border, which in **d** was increased in intensity and width in discs lacking Fu kinase activity (*fu*^{*mH63*}), and in **f** and **h**, was decreased in intensity in discs lacking *Su(fu)* (**f**) or both *Su(fu)* and Fu kinase activity (**h**). **c, g**, *ci* RNA (blue) levels in wild-type discs (**c**) were unaltered in the absence of *Su(fu)* (**g**).

that is otherwise antagonized by *Su(fu)*; this feature of Hh signalling is reproduced by *ptc* mutation but not by *PKA* mutation.

Hh signalling thus elicits several changes that are required to convert Ci into an effective transcriptional activator. Hh spares Ci-155 from *PKA*- and *Slimb*-dependent proteolysis to Ci-75 (refs 1, 18), perhaps by modifying the phosphorylation status of Ci (ref. 25), and promotes dissociation of the Ci-155 complex from microtubules⁸. We propose that in wing discs some of this 'primed' Ci-155 is not associated with *Su(fu)* and can activate *dpp* and *ptc* but not anterior *en* expression. Most of the primed Ci-155 in wing discs and perhaps all of the primed Ci-155 in embryos is inactive while it is in complex with *Su(fu)* and signalling by Hh and Fu kinase are necessary for it to become a transcriptional activator. This active form of Ci appears to be unstable and so is not detectable in the nuclei of cells responding to Hh.

The lower levels of Ci-155 that are found in wing discs close to the source of Hh compared with those in more distant regions of the Hh-signalling domain^{12,26} may be explained by our model if more Hh is required to stimulate the Fu-kinase-dependent step in Ci activation than to protect Ci-155 from proteolytic degradation to Ci-75. This dosage dependence may account for the restricted range of anterior *en* induction relative to *dpp* and *ptc* in wing discs^{11,12,26} and the single-cell range of Hh signalling in the embryonic ectoderm. □

Methods

Genetic analysis. Most wing disc clones were generated by FRT-mediated recombination in first instar larvae using the following alleles: *DC0*^{B3} (null allele of the principal PKA catalytic subunit)²⁷, *ptc*^{S2} (strong hypomorph), *fu*^{*mH63*} (strong hypomorphic class I allele)^{19,20}, *sno*^{11X43} (null allele)²⁴, *slimb*¹ (hypomorphic) and *slimb*² (probably null)¹⁸. *P[*fu*⁺]* rescue transgenes²⁸ on 2L and 2R were used to generate *ptc fu*, *PKA fu* and *sno fu* double-mutant clones. *Su(fu)*^{LP} is null²⁹ and *Df(4)M62f* eliminates both *ci* and *pangolin* gene activity³⁰. **Analysis of wings.** Wings were mounted in Canada balsam/methyl salicylate (1:1) and clones marked by *stc* or *sha* and *y* at the margin were characterized for

each genotype only if they occupied congruent dorsal and ventral surfaces and extended from the wing margin up to or near the vein 3–4 crossvein.

Imaginal disc and embryo staining. Standard protocols were followed, using the anti-Ci antibody 2A1 and the anti-En antibody 4D9 with Vectastain Elite ABC kits and horseradish peroxidase staining for comparing the staining intensity of samples processed in parallel, or immunofluorescent secondary antibodies to confirm the location of clones.

Western blots. Approximately 50 μ l packed stage-11–12 embryos were homogenized in 150 μ l RIPA buffer before adding an equal volume of loading buffer to give ~ 0.4 mg ml⁻¹ protein. Generally, 100 wing discs were dissected into 100 μ l loading buffer and vortexed to homogenize them. Blots were probed successively with antibodies raised against the Ci zinc-finger domain (from P. W. Ingham), the Ci C-terminal half¹⁷, and the RII subunit of PKA or α -tubulin, and visualized using horseradish peroxidase-coupled secondary antibody and a chemiluminescent reagent (Pierce).

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Local calcium signalling by inositol-1,4,5-trisphosphate in Purkinje cell dendrites

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The second messenger inositol-1,4,5-trisphosphate (InsP₃) releases Ca²⁺ from intracellular Ca²⁺ stores by activating specific receptors on the membranes of these stores¹. In many cells, InsP₃ is a global signalling molecule that liberates Ca²⁺ throughout the cytoplasm^{1,2}. However, in neurons the situation might be different^{3,4}, because synaptic activity may produce InsP₃ at discrete locations. Here we characterize InsP₃ signalling in

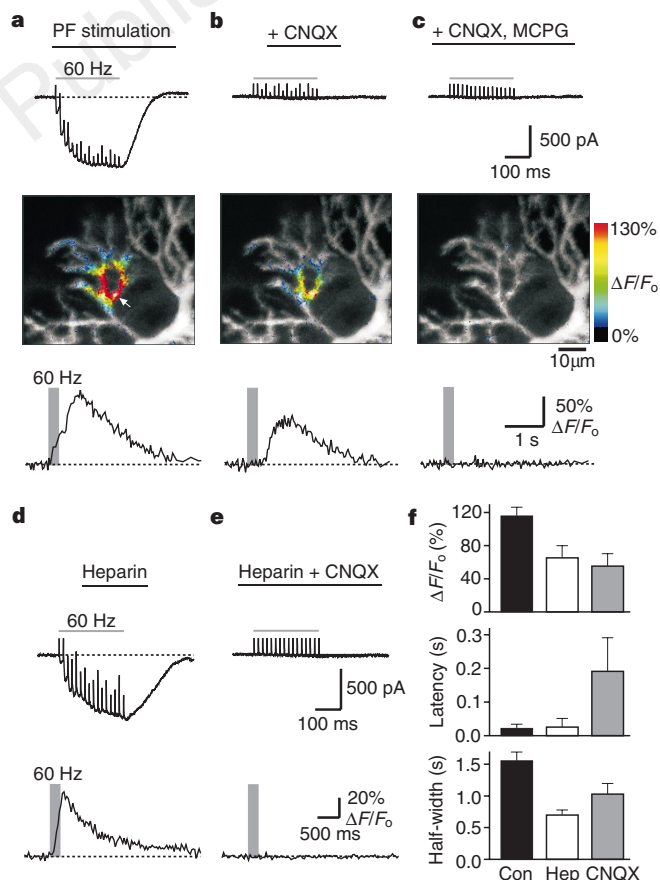


Figure 1 Components of postsynaptic Ca²⁺ signalling during parallel-fibre (PF) synaptic activity. **a–c**, Purkinje cell responses to repetitive PF stimulation (250 ms, 60 Hz; indicated by bars or shaded boxes) in **a**, control conditions, **b**, CNQX (15 μ M), and **c**, CNQX (15 μ M) plus MCPG (1 mM). Top, electrical responses. Middle, Ca²⁺ signals (pseudocolour), measured at the peak of the response, superimposed on a reconstructed image of the Purkinje cell (greyscale). Bottom, time course of Ca²⁺ signals, measured in a 5- μ m dendritic segment (arrow in **a**). **d**, **e**, Responses to repetitive PF stimulation (250 ms, 60 Hz) in Purkinje cells dialysed with heparin (50 μ g ml⁻¹) in **d**, the absence or **e**, the presence of CNQX (15 μ M). Top, electrical responses. Bottom, time course of the Ca²⁺ signal. **f**, Properties of Ca²⁺ signals produced by PF stimulation (250 ms, 60 Hz) in control conditions ($n = 4$), in cells dialysed with heparin ($n = 3$), and in cells treated with CNQX ($n = 4$).